CONSULTANCY / SOLUTION / DELIVERY

Assembly IT Infrastructure Designed for production.

End-to-end network consulting, structured cabling, fiber backbone, LAN, Wi-Fi 7, CCTV, PA and rack integration for WAGO’s first-floor assembly infrastructure in Vadodara.

PROJECT PROMISE

Plan precisely. Integrate cleanly. Deliver with evidence.



EXECUTIVE SNAPSHOT

A production-ready foundation built from design to delivery.

Documented project scale based on Purchase Order 40227067 and its technical annexure.

2	**46**	**4,000 m**	**750 m**	**5**
22U floor racks	**data outlets**	**Cat7 cable**	**fiber routes**	**managed switches**
Dual infrastructure zones	Cat6A shielded termination	Assembly-table connectivity	OM4 + single-mode backbone	Core, access and PoE

INTEGRATED SCOPE

LAN + fiber + Wi-Fi 7 + CCTV + PA + pathways

One accountable design and delivery model for the first-floor assembly environment - from rack and route planning through configuration, labeling and handover.

BUSINESS REQUIREMENT

Connect the assembly floor without compromising order, resilience or serviceability.

The requirement combined high-density data points with fiber uplinks, wireless coverage, surveillance and plant communication.

01

Production connectivity

Reliable connections for assembly tables and equipment across a large floor plate.

02

Structured pathways

Segregated, supported routes using tray, conduit and protected drops.

03

Scalable distribution

Rack-based switching, patching and fiber termination designed for maintainability.

04

Integrated services

Wi-Fi 7, entry CCTV and PA endpoints delivered within the same infrastructure plan.

DESIGN INTENT A clean, labeled and testable infrastructure that can grow with production demand.

CONSULTING APPROACH

The solution began with engineering decisions - not a product list.

A controlled path from requirement capture to validated implementation.

01

DISCOVER

Floor use, endpoint count, rack zones and service requirements.

OUTPUT

02

DESIGN

Topology, pathways, rack layout, media selection and BoQ.

OUTPUT

03

COORDINATE

OEM compatibility, site readiness, safety and execution sequencing.

OUTPUT

04

DELIVER

Supply, routing, installation, splicing, dressing and configuration.

OUTPUT

05

VALIDATE

Label checks, link testing, device status and visual quality review.

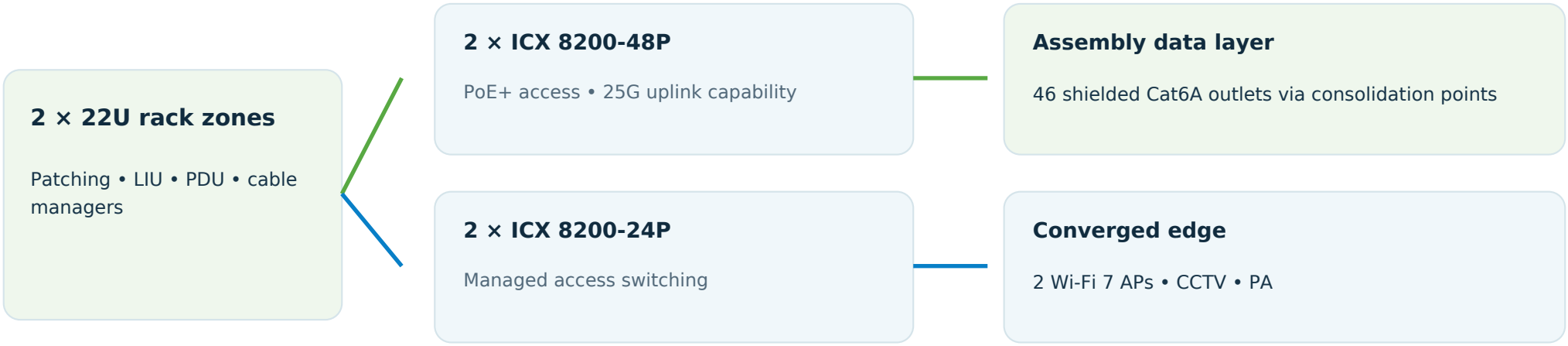
OUTPUT

The consultancy converts site constraints into buildable drawings, quantities, installation standards and acceptance evidence.

SOLUTION DRAWING 01

Logical network architecture for converged plant services.

Illustrative topology based on the documented equipment and media schedule.



10G OPTICS / UPLINKS

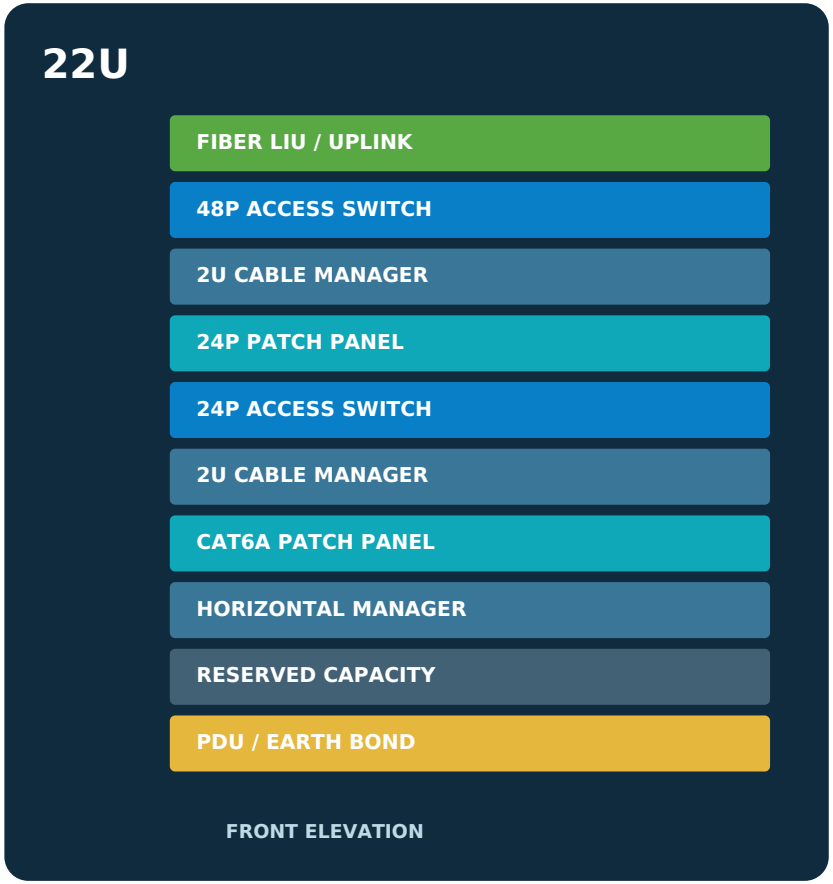
BACKBONE

600 m OM4 multimode + 150 m single-mode fiber • LC termination • 72 documented splices

SOLUTION DRAWING 02

Rack elevation designed for access, airflow and serviceability.

Representative 22U arrangement; final position follows the as-built rack schedule.



01 Fiber first

Accessible LIU placement and protected patch leads.

02 Managed dressing

Horizontal managers control bend radius and port visibility.

03 Growth space

Reserved U-space avoids rebuilding for incremental expansion.

04 Safe power

Rack PDU, mechanical lock, ventilation and earth bonding.

SOLUTION DRAWING 03

A disciplined physical route from rack to assembly point.

Pathway design protects cable performance and keeps future maintenance practical.

CEILING DISTRIBUTION	300 m perforated cable tray • supports • bends • T branches
BACKBONE MEDIA	OM4 / single-mode fiber and Cat7 horizontal distribution
CONTROLLED DROP	32 mm conduit, flexible conduit or HDPE protection
WORK AREA	Consolidation point → shielded Cat6A → assembly outlet

SEGREGATED | SUPPORTED | LABELED | TESTABLE | EXPANDABLE

INTEGRATED SOLUTION

One coordinated scope across nine technology workstreams.

Supply and services were engineered as one system - not disconnected packages.

RACKS

2 × 22U, PDU and cable management

LAN

2 × 48P + 2 × 24P managed PoE switches

DATA

46 shielded Cat6A outlets and consolidation points

COPPER

4,000 m Cat7 + rack/user patching

FIBER

600 m OM4 + 150 m SM + LIUs and splicing

WI-FI

2 quad-radio Wi-Fi 7 access points

CCTV

5MP IR dome + managed PoE edge switch

PA

5 ceiling + 5 wall speakers and cabling

PATHWAYS

300 m tray + conduit / flexible / HDPE

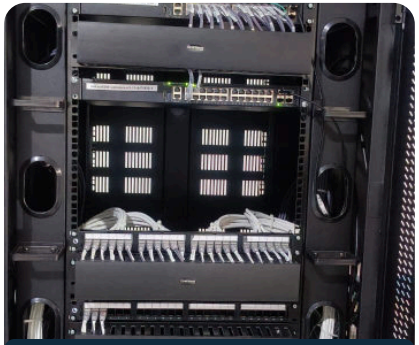
DELIVERY EVIDENCE

From drawing to disciplined rack build.

Actual site photographs show installed racks, patching, fiber termination, switching and cable dressing.



RACK BUILD



PATCHING



STRUCTURED DRESSING

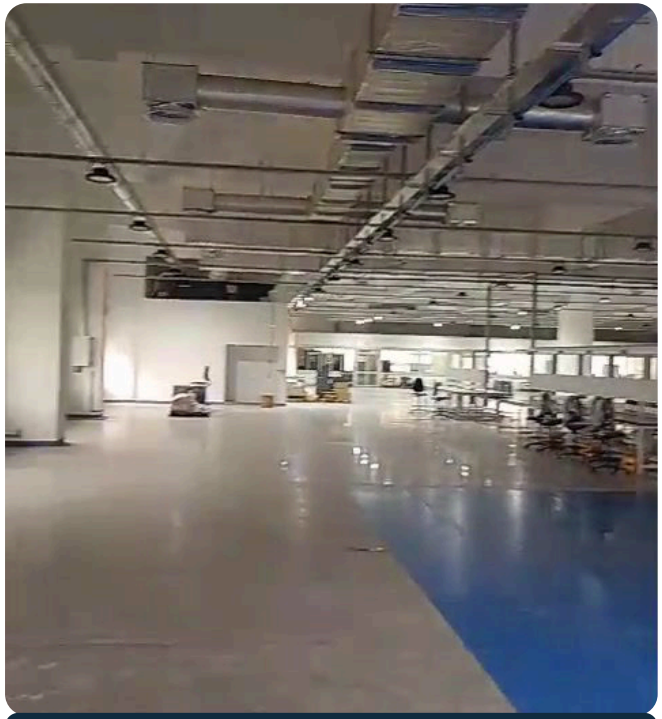


FIBER + ACCESS

SITE EXECUTION

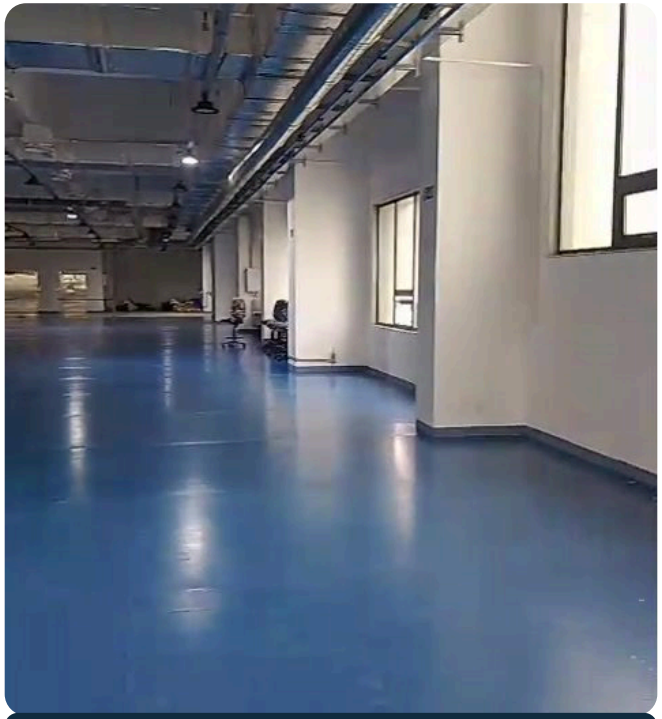
Pathways, racks and final integration executed inside the assembly environment.

The implementation sequence was coordinated around site access and production-floor readiness.



01 / POSITION

Rack zone and route interface



02 / ROUTE

Overhead tray and protected drops



03 / INTEGRATE

Termination, configuration and dress

QUALITY ASSURANCE

A professional finish is measurable - not only visible.

Recommended acceptance evidence for infrastructure handover and lifecycle support.

Physical QA

01

Rack anchoring, earth bond, ventilation, bend radius and tray support.

Copper QA

02

Point labels, continuity and certification records for shielded links.

Fiber QA

03

Core mapping, splice records, polarity and optical-loss results.

Active QA

04

Switch, uplink, PoE, AP and camera link / power verification.

Documentation

05

As-built topology, rack schedule, port map and asset register.

Handover

06

Open-item closure, configuration backup and support escalation path.

HANDOVER STANDARD **Installed • labeled • tested • documented • support-ready**

CASE STUDY CONCLUSION

Consulted with purpose. Designed with discipline. Delivered with evidence.

D-TECH translated a multi-technology assembly-floor requirement into one integrated infrastructure solution - combining pathways, copper, fiber, switching, Wi-Fi, CCTV, PA and professional rack integration.



D-TECH SOLUTION INTEGRATORS PVT. LTD.

Consultancy • solution architecture • supply • implementation • lifecycle support